

A Three-Site Variational Hénon Ring: An Exact Fourier-Mode Period-Two Witness

Route-A finite certificate C111

Abstract

We give a small exact certificate for a genuinely three-site variational Hénon lattice. A triangular coupling graph, with $a = 7$ and $\kappa = 1/5$, yields a polynomial exact-symplectic map over $\mathbb{Q}$. Two synchronous fixed points and one primitive synchronous period-two orbit are verified by independent rational arithmetic. The two-step Jacobian splits into one longitudinal and two transverse Laplacian modes, giving the factorisation $(1 + 7z + z^2)(1 + 106z/25 + z^2)^2$ for the finite monodromy polynomial. This is deliberately a low-period witness; no global zeta or arithmetic claim is made.

1 Scope and model

This paper records package C111 in the exploratory Route-A line. The claim is an auditable finite calculation, not a global zeta theorem. Let $q, p \in \mathbb{Q}^3$ and let

$$L = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}, \quad U(q) = \sum_{i=1}^3 \left(\frac{7}{2} q_i^2 - \frac{1}{3} q_i^3 \right) - \frac{1}{10} \sum_{\{i,j\}} (q_i - q_j)^2.$$

The sum has the three edges of the 3-cycle. The map is

$$F(q, p) = (G(q) - p, q), \quad G(q) = \nabla U(q). \tag{1}$$

All computations use exact fractions. The evidence file and scripts are listed in the reproducibility paragraph below.

2 Exact variational identities

The Hessian and Jacobian are

$$H(q) = \text{diag}(7 - 2q_i) - \frac{1}{5} L, \quad J(q) = \begin{pmatrix} H(q) & -I_3 \\ I_3 & 0 \end{pmatrix}.$$

With $\Omega = \begin{pmatrix} 0 & I_3 \\ -I_3 & 0 \end{pmatrix}$, direct multiplication gives $J(q)^\top \Omega J(q) = \Omega$ and $\det J(q) = 1$. The coordinate swap $R(q, p) = (p, q)$ is an involutory reversor: $RFR = F^{-1}$. For the canonical one-form $\lambda = q \cdot dp$ one has the exact primitive identity

$$F^* \lambda - \lambda = d(U(q) - p \cdot q).$$

These are polynomial identities, checked both by the producer/checker and by the independent SymPy script.

3 Certified orbit witnesses

The synchronous subspace is invariant because $L(1, 1, 1)^\top = 0$. The two fixed states and the period-two states are

$$\begin{aligned} P_0 &= ((0, 0, 0), (0, 0, 0)), \\ P_5 &= ((5, 5, 5), (5, 5, 5)), \\ C_3 &= ((3, 3, 3), (6, 6, 6)), \quad C_6 = ((6, 6, 6), (3, 3, 3)). \end{aligned}$$

Substitution into (1) gives $F(P_i) = P_i$ and $F(C_3) = C_6$, $F(C_6) = C_3$. Since $C_3 \neq C_6$, this is a primitive period-two witness. No completeness statement is made about other fixed or periodic points.

Table 1: Exact two-step mode data on the period-two witness.

mode	Laplacian eigenvalue	(h_3, h_6)	trace
longitudinal	0	$(1, -5)$	-7
transverse 1	3	$(2/5, -28/5)$	$-106/25$
transverse 2	3	$(2/5, -28/5)$	$-106/25$

4 Longitudinal and transverse modes

The Laplacian eigenvalues are 0, 3, 3. At a synchronous point with scalar coordinate s , the Hessian eigenvalue in mode ℓ is

$$h_s(\ell) = 7 - 2s - \frac{1}{5}\ell.$$

Thus the two points of the cycle have mode pairs

$$(h_3(0), h_6(0)) = (1, -5), \quad (h_3(3), h_6(3)) = (2/5, -28/5).$$

For a scalar mode, the two-step matrix $\begin{pmatrix} h_6 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} h_3 & -1 \\ 1 & 0 \end{pmatrix}$ has determinant one and trace $h_3h_6 - 2$. Therefore the mode traces are

$$t_0 = -7, \quad t_{\perp} = -\frac{106}{25} \quad (\text{multiplicity two}).$$

The direct 6×6 monodromy and the mode reconstruction agree exactly:

$$\det(I - zM) = (1 - t_0z + z^2)(1 - t_{\perp}z + z^2)^2 = 1 + \frac{387}{25}z + \frac{50211}{625}z^2 + \frac{98002}{625}z^3 + \frac{50211}{625}z^4 + \frac{387}{25}z^5 + z^6. \quad (2)$$

As a finite coupling control, setting $\kappa = 0$ at the same synchronous states gives the coefficient vector

$$(1, 21, 150, 385, 150, 21, 1),$$

instead of the coefficients in (2); the trace changes by $138/25$. This comparison isolates a coupling effect without interpreting either polynomial as a Fredholm determinant.

5 Reproducibility and limits

From the package directory, run

```
python3 code/c111_three_site_producer.py
python3 code/c111_three_site_checker.py
python3 code/c111_sympy_crosscheck.py
python3 code/c111_replay.py
python3 code/c111_mutation.py
```

The canonical evidence is `results/c111_three_site_evidence.json`; the mutation audit rejects all twelve semantic edits. The release manifest records hashes for every source, result, and paper artifact.

The route verdict is `A1 = A1_WEAK` (certified low-period witnesses), `A2 = A2_FAIL` (operator owner open), `A3` not addressed, and `A4` failed; overall status is `ROUTE_A_EXPLORATORY`. In particular, this package does not claim a complete primitive-orbit atlas, a global transfer or Fredholm operator, analytic continuation, Euler factors, root numbers, automorphy, or a Hilbert–Pólya operator. Its scope literal is `NO_BAD_EULER_OR_ROOT_NUMBER`.